TEAM COLLABORATION & MEETING LOGS

Regular group meetings were conducted to ensure systematic progress and collaborative decision-making throughout the project. These meetings allowed the team to discuss different stages of the project including dataset understanding, preprocessing strategies, exploratory data analysis, machine learning model development, evaluation methods, and final documentation preparation.

Meetings were held on the following dates:

- **6th March 2026**
- **7th March 2026**
- **9th March 2026**
- **10th March 2026**
- **12th March 2026**

During these meetings, the team discussed **dataset selection, preprocessing strategies, exploratory data analysis (EDA), machine learning model implementation, evaluation metrics such as Mean Time to Detect (MTTD), polling frequency analysis, testing procedures, and the preparation of the final project report and documentation**. Each meeting helped ensure that tasks were distributed properly among team members and that the project progressed smoothly.

Meeting 1 — 6th March 2026

Topics Discussed: Project overview, dataset understanding, problem statement analysis, planning project workflow, division of work among team members.

The first team meeting focused on understanding the **project objectives and problem statement**. The team discussed the importance of detecting faults in smart streetlight systems using IoT sensor data. The goal of the project was explained clearly, which is to build a machine learning model capable of classifying streetlight status into **operational or faulty categories**.

During this meeting, the dataset structure was introduced and examined. The team reviewed the dataset attributes such as **power consumption, voltage levels, current fluctuations, temperature, and environmental conditions**. The **fault_type variable** was identified as the target variable representing whether a streetlight is functioning normally or experiencing a fault.

The team also discussed the **overall workflow of the project**, including dataset preprocessing, exploratory data analysis, machine learning model training, and evaluation. Responsibilities were distributed among team members to ensure efficient collaboration. Some members were assigned tasks related to dataset preparation, while others focused on visualization, machine learning implementation, and documentation.

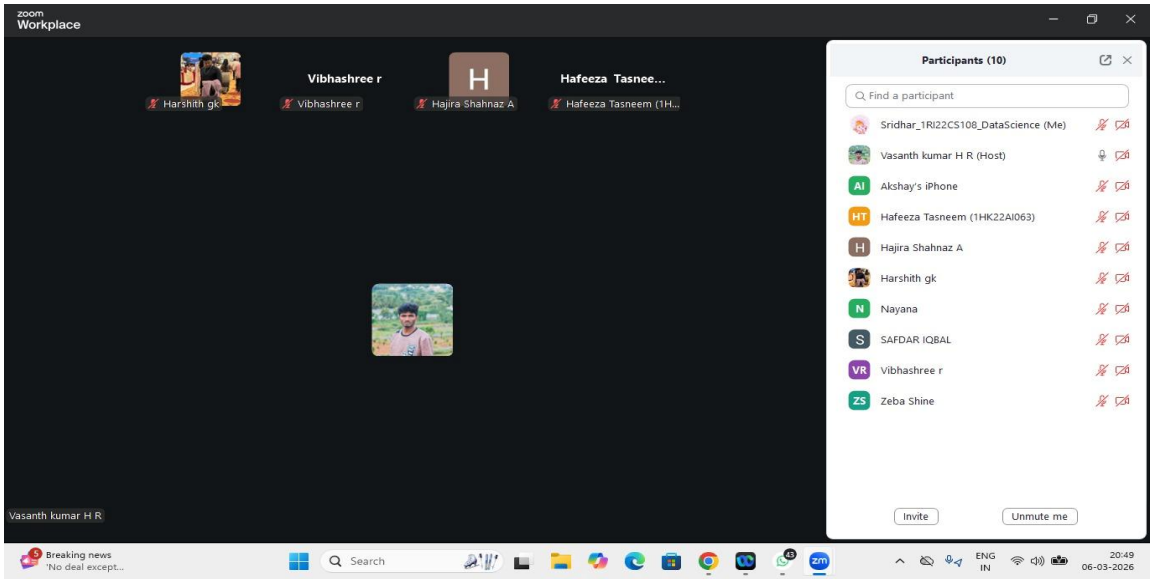

Figure A: Team meeting discussing project overview and dataset structure.

Meeting 2 — 7th March 2026

Topics Discussed: Dataset preprocessing, dataset inspection, cleaning dataset, preparing dataset for analysis.

The second meeting focused on **data preprocessing and dataset preparation**. The dataset was loaded into the Python environment using the **Pandas library**, and several commands such as df.head(), df.info(), and df.describe() were used to examine the dataset.

The team checked for **missing values, incorrect data types, and inconsistencies in the dataset**. Since the dataset contained complete records, only minimal preprocessing was required. However, the team discussed potential techniques such as **feature normalization and data transformation** if needed during the machine learning stage.

Team members responsible for data handling ensured that the dataset was properly structured for further analysis.

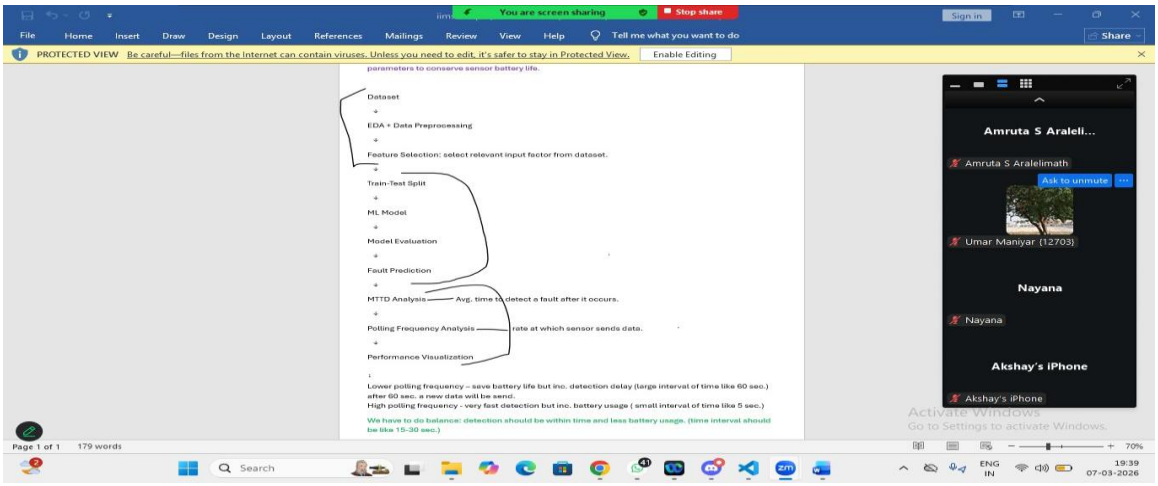

Figure B: Team discussion on dataset preprocessing and data inspection.

Meeting 3 — 9th March 2026

Topics Discussed: Exploratory Data Analysis (EDA), visualization techniques, understanding feature relationships.

The third meeting focused on **Exploratory Data Analysis (EDA)** to better understand the patterns and relationships within the dataset. Various visualization techniques were used to analyze the sensor data.

The team generated **histograms to observe feature distributions**, **boxplots to detect outliers**, and **correlation heatmaps to examine relationships between variables**. A count plot was also used to analyze the distribution of the target variable **fault_type**, which indicates the number of operational and faulty streetlights.

These visualizations helped identify patterns in the data, such as variations in power consumption and voltage levels between operational and faulty streetlights. The insights obtained from EDA helped guide the feature selection process and prepare the dataset for machine learning modeling.

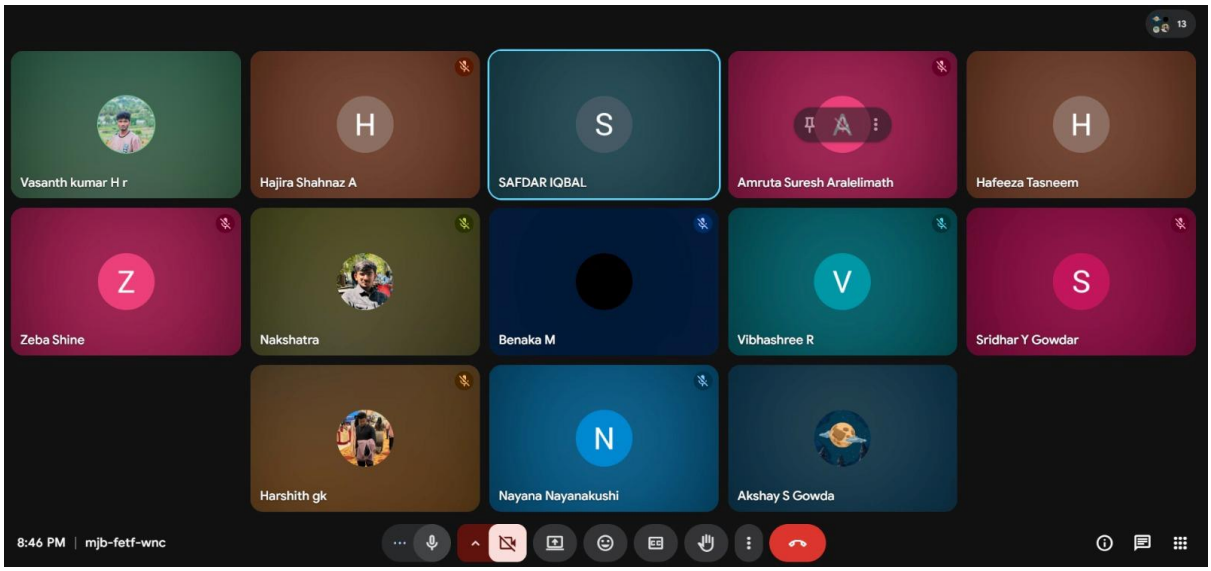

Figure C: Team meeting discussing EDA visualizations and insights.

Meeting 4 — 10th March 2026

Topics Discussed: Machine learning model implementation, feature selection, training and testing datasets, Mean Time To Detect (MTTD) concept.

The fourth meeting focused on the **implementation of machine learning algorithms** for streetlight fault detection. The team finalized the input features and separated them from the target variable.

The dataset was then divided into **training and testing datasets** using the train-test split method. This allowed the model to learn from historical data and evaluate its performance on unseen data.

The team also discussed the **Mean Time To Detect (MTTD)** metric, which measures the average time taken by the system to detect faults after they occur. Understanding MTTD is important for evaluating the efficiency of smart monitoring systems.

Different machine learning algorithms were discussed, including **Logistic Regression, Decision Tree, and Random Forest**. These models were evaluated to determine which algorithm provided the best performance for the dataset.

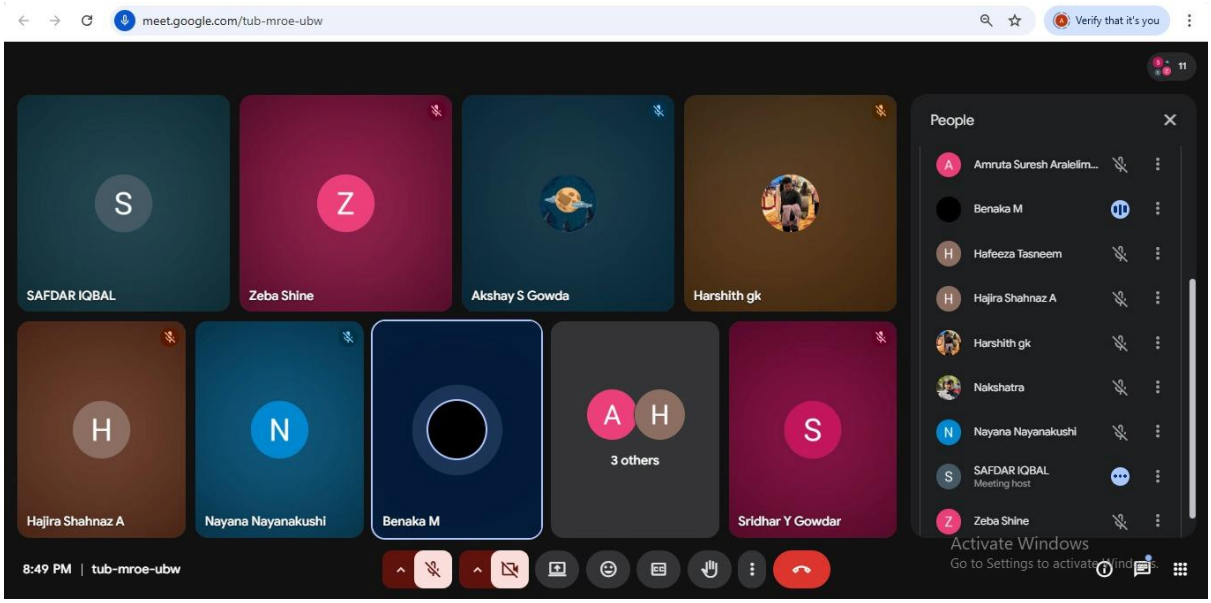

Figure D: Team meeting discussing machine learning models and evaluation metrics.

Meeting 5 — 12th March 2026

Topics Discussed: Polling frequency analysis, testing the model, final project completion, documentation preparation and review.

The final meeting focused on **testing the developed machine learning model and reviewing the project results**. The team analyzed how accurately the model could detect faulty streetlights based on sensor data.

During this meeting, the concept of **polling frequency analysis** was discussed. Polling frequency determines how often IoT sensors send data to the monitoring system. The team discussed how higher polling frequency allows faster fault detection but may increase sensor battery consumption.

The team also reviewed the **project results, visualizations, and evaluation metrics**. Finally, the documentation and project report were reviewed to ensure that all sections were complete and properly structured.

This meeting concluded with the **final completion of the project and preparation for submission and presentation**.

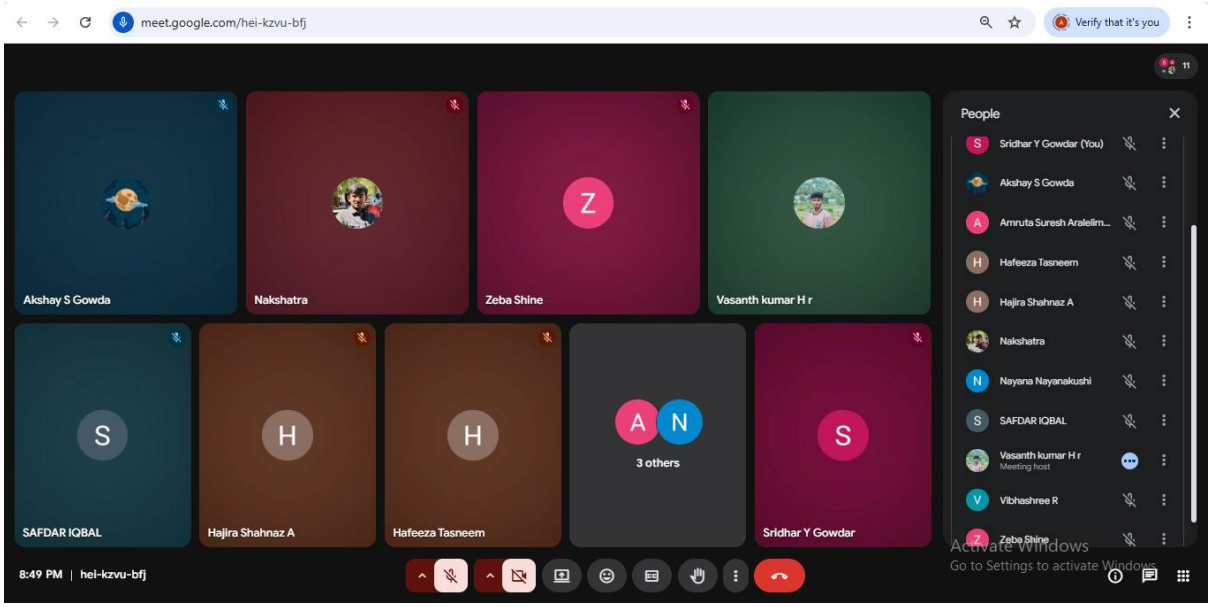

Figure E: Final team meeting reviewing project results and documentation.